TECHNICAL REPORT

Problem Statement:

Modern telecom networks generate a massive amount of telemetry data from thousands of cells towers every few minutes. Network operators continuously monitor Key Performance Indicators (KPIs) such as RSRP, SINR, Latency, Throughput, Packet Loss, and Connected Users to ensure high network availability and service quality. Manual monitoring of such large-scale data is inefficient and may delay fault detection.

The objective of this project is to develop an Artificial Intelligence for IT Operations (AIOps) system that automatically analyzes telecom KPI data, detects anomalies, forecasts future traffic, predicts Quality of Service (QoS), performs Root Cause Analysis (RCA), and provides a real-time Streamlit dashboard for network monitoring.

Dataset Description:

The dataset contains 30 days of telecom network KPI data collected at 5-minute intervals from multiple cell sites.

Dataset Features:

Feature	Description
Timestamp	Date and Time of KPI measurement
Cell_ID	Cell Tower Identifier
RSRP	Reference Signal Received Power (dBm)
SINR	Signal to interference plus Noise Ratio (dB)
Latency	Network Delay (ms)
Throughput	Data Transfer Rate (Mbps)
Packet_Loss	Percentage of packet loss
Connected_Users	Number of active users
Qos_Label	Network Quality (Good, Fair, Poor)
KPI_Label	Network Status (Normal/Warning/Critical)

Dataset Characteristics:

- Duration: 30 days
- Sampling Interval: 5 minutes
- Multiple cell towers
- Contains both normal and abnormal network behavior
- Suitable for time-series forecasting and anomaly detection

Time-Series Analysis

Time-series analysis was performed to understand the behavior of telecom KPIs over time.

Preprocessing Steps:

- Converted timestamp into datetime format
- Sorted records chronologically
- Checked missing timestamps
- Verified missing KPI values
- Generated daily and weekly KPI trends

KPI Trends

- Throughput decreases during heavy traffic periods.
- Latency increases during congestion
- Packet loss rises during network failures
- SINR decreases because of radio interference
- RSRP varies depending on signal coverage

Outcome

Time-series visualization helped identify daily traffic peaks and long-term network performance trends.

Isolation Forest Results

Isolation Forest was used as an unsupervised machine learning algorithm to detect abnormal KPI behavior.

Steps Perform

- Selected KPI features
- Standardized the data
- Trained the Isolation Forest model
- Calculated anomaly scores
- Exported detected anomalies

Results

- Successfully detected abnormal KPI records
- Highlighted sudden drops in Throughput
- Identified unusually high latency and packet loss

- Generated anomaly reports for further investigation

Advantages

- No labeled data required
- Fast anomaly detection
- Suitable for large telecom datasets

LSTM Model Summary

A Long Short-Term Memory (LSTM) neural network was implemented for sequential anomaly detection and network traffic forecasting.

Architecture

- Input Layer
- Two LSTM layers
- Dropout layer
- Fully connected output layer

Hyperparameters

Parameter	Value
Sequence Length	20
Hidden Units	64
Number of Layers	2
Batch Size	64
Epochs	20
Optimizer	Adam
Loss Function	Mean Squared Error

Results

The LSTM successfully learned normal telecom traffic patterns. Prediction errors were used to identify anomalous network behavior.

Traffic Forecasting Results

An LSTM forecasting model was developed to predict future throughput.

Forecasting Duration

- Next 24 hours
- Next 7 days

Evolution Metrics

- Mean Absolute Error
- Root Mean Square Error
- Mean Absolute Percentage Error

Observations

- The model accurately predicted short-term traffic trends
- Peak traffic hours were successfully forecasted
- Forecasting enables proactive resource allocation and congestion management

Root Cause Analysis (RCA)

Each detected anomaly was analyzed to identify its probable cause using telecom KPI thresholds.

RCA Rules

Observation	**Root Cause**	**Recommended Action**
Low RSRP	Poor Radio Coverage	Optimize antenna tilt
Low SINR	High Interference	Review neighboring cells
High Latency	Core Network Congestion	Check UPF utilization
High Packet Loss	Backhaul Issues	Inspect transport network
Low Throughput	Heavy Traffic Load	Enable load balancing
High Connected Users	Cell Overload	Deploy additional small cells

Outcome

The RCA module automatically generated probable causes and suggested corrective actions for detected anomalies

QoS Prediction Results

A Random Forest classifier was developed to predict network quality.

Classes

- Good
- Fair
- Poor

Input Features

- RSRP

- SINR
- Latency
- Packet Loss
- Throughput

Evolution Metrics

- Accuracy
- Precision
- F1 Score
- Recall

Results

The model achieved high classification performance and accurately predicted network quality using telecom KPIs. The predicted QoS labels assist network operators in maintaining service reliability.

Conclusions

This project successfully demonstrated the application of Artificial Intelligence for telecom network monitoring.

The developed AIOps framework integrates time-series analysis, anomaly detection, traffic forecasting, QoS prediction, Root Cause Analysis, and an interactive Streamlit dashboard into a single solution. The system enables proactive fault detection, minimizes manual monitoring efforts, and supports faster decision making for network administrators.

The implementation shows how ML and DL techniques can significantly improve telecom network performance and operational efficiency.

Future Improvements

- Real-time KPI streaming using Apache Kafka
- LSTM Autoencoder for more accurate anomaly detection
- Transformer-based time-series forecasting models
- Reinforcement Learning for self-optimizing setworks.
- Integration with 5G Network Slicing
- Explainable AI (XAI) for model interpretability

- Automated Root Cause Analysis using Graph Neural Network
- Cloud deployment using Docker and Kubernetes
- Real-time alert notifications via email or SMS
- Integration with telecom OSS/BSS platforms for automated fault management